CRYPTO HARD

Nonce Nursery

Overview & Vulnerability

Authored by 5h1kh4r. Files: `chall.py` and `data.json`. The challenge presents 11 ECDSA signatures on curve `secp256k1`. While 6 nonces are generated uniformly at random, a contiguous window of 5 nonces ($k_0 \dots k_4$) is generated using an affine Linear Congruential Generator (LCG): $k_{j+1} \equiv A \cdot k_j + B \pmod N$
The starting index of this length-5 sequence is unknown, but with only 11 signatures, there are only 7 possible window offsets ($start \in [0, 6]$).

Nonce Representation in Private Key d

Inspecting the signature records from `data.json`:

```
$ head -n 25 data.json
```



Signature Records Inspection

In ECDSA, each signature (r_i, s_i) on message hash h_i relates to the private key d via: $k_i \equiv s_i^{-1} h_i + (s_i^{-1} r_i) \cdot d \equiv \alpha_i + \beta_i \cdot d \pmod N$ Where α_i and β_i are publicly known constants for each signature.

Algebraic Elimination of A and B

For any 5-nonce LCG sequence $k_0 \dots k_4$: 1. **Cancel constant B:** Taking first-order differences $\Delta_j = k_{j+1} - k_j$ gives a geometric sequence: $\Delta_1 \equiv A \cdot \Delta_0$ $\Delta_2 \equiv A \cdot \Delta_1 = A^2 \cdot \Delta_0$ $\Delta_3 \equiv A \cdot \Delta_2 = A^3 \cdot \Delta_0$ 2. **Cancel multiplier A:** Cross-multiplying successive terms removes A: $\Delta_2 \cdot \Delta_0 - \Delta_1^2 \equiv 0 \pmod N$ $\Delta_3 \cdot \Delta_1 - \Delta_2^2 \equiv 0 \pmod N$ 3. **Linearize d:** Substituting $\Delta_j = u_j + v_j \cdot d$ yields two quadratic polynomials in d: $P_1(d) = a_1 d^2 + b_1 d + c_1 \equiv 0 \pmod N$ $P_2(d) = a_2 d^2 + b_2 d + c_2 \equiv 0 \pmod N$ Taking the linear combination $a_2 P_1(d) - a_1 P_2(d)$ cancels the d^2 term, directly solving for d: $d \equiv -(a_2 c_1 - a_1 c_2) \cdot (a_2 b_1 - a_1 b_2)^{-1} \pmod N$

Solver Implementation

Solution Script: `Nonce_Nursery.py` (included in attached solution archive)

Output

Running the algebraic solver:

```
$ python3 solve.py
```

```
~/SecLeaf writeups/repo_clone/SecLeafCTF/CRYPTO/Nonce_Nursery main*
❯ python3 solve.py
[+] Found affine sequence at window indices 3..7
[+] Recovered private key d: 11226524650944399179350180557504702582548594673059420756561227687536795622371O
[+] Flag: SecLeaf{4ff1n3_n0nc3s_b3nd_th3_curv3}

~/SecLeaf writeups/repo_clone/SecLeafCTF/CRYPTO/Nonce_Nursery main*
❯
```

Solver Execution Proof

Solution Script: Nonce_Nursery.py (included in attached solution archive)

Flag:

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RECOVERED FLAG **SecLeaf{4ff1n3_n0nc3s_b3nd_th3_curv3}**